



# **Rungta College of Engineering and Technology Bhilai (CG)**

**Department of CSE- Cyber Security**

**Project Proposal Presentation**

## **OSINT TOOLKIT DASHBOARD**

***Cyber Security***

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# OSINT Toolkit Dashboard

Cybersecurity Reconnaissance & Domain Intelligence  
Platform

WHOIS + DNS Intelligence Dashboard

Built using HTML, CSS, JavaScript, Node.js and Express.js

# Introduction

OSINT stands for Open Source Intelligence.  
The project analyzes publicly available domain information.

Integrates WHOIS, DNS and Social OSINT reconnaissance tools into one dashboard.

Designed for cybersecurity learning and reconnaissance analysis.

# Problem Statement

Traditional OSINT tools are difficult for beginners.

Most tools focus on only one reconnaissance capability.

This project creates a centralized visual dashboard for intelligence gathering.

# Objectives

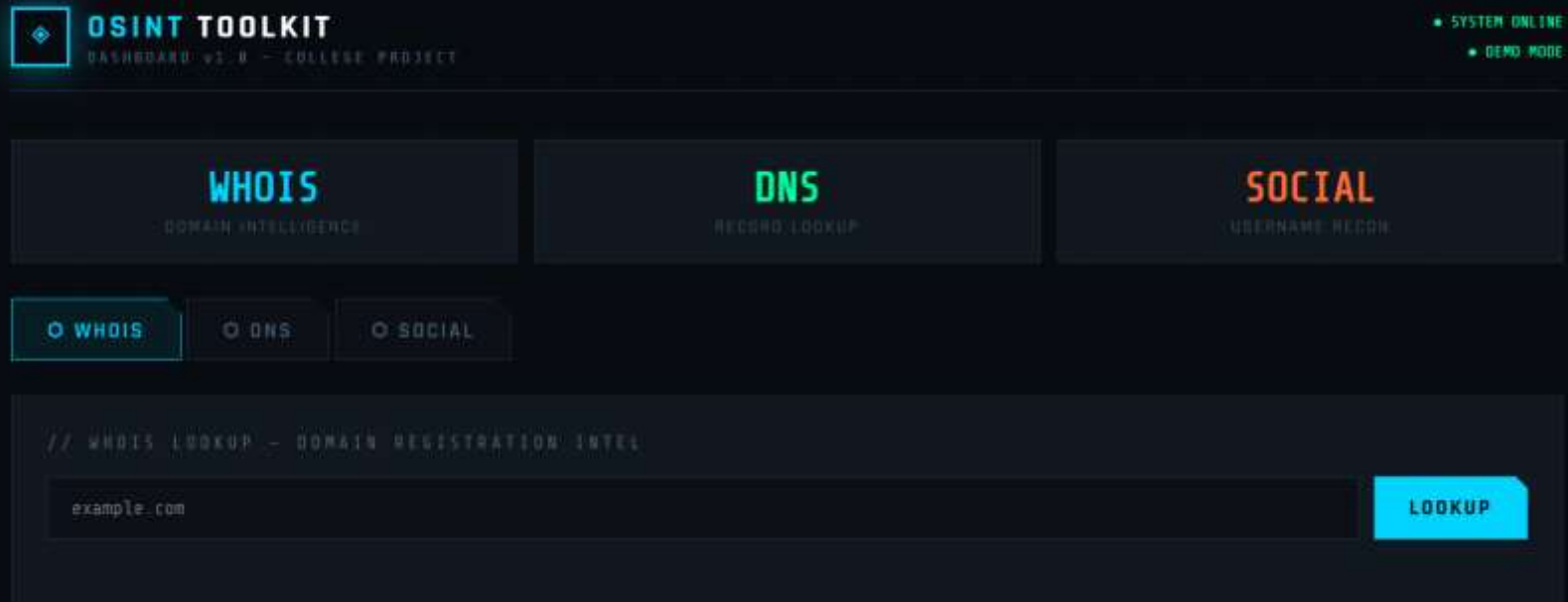
Learn frontend/backend integration.

Implement WHOIS and DNS lookup modules.

Understand API communication using Fetch API.

Build a responsive cybersecurity dashboard interface.

# Project Overview



Unified dashboard for domain reconnaissance.  
Supports WHOIS analysis and DNS record lookups.  
Frontend communicates with backend APIs in real time.

# Technologies Used

Frontend: HTML, CSS, JavaScript

Backend: Node.js, Express.js

Libraries: whois-json, dns, cors

Development Tools: VS Code, Live Server

Communication: Fetch API + JSON

# WHOIS Module

WHOIS

DOMAIN INTELLIGENCE

DNS

RECORD LOOKUP

SOCIAL

USERNAME RECON

WHOIS

DNS

SOCIAL

// WHOIS LOOKUP - DOMAIN REGISTRATION INTEL

google.com

LOOKUP

DOMAIN  
google.com

REGISTRAR  
MarkMonitor, Inc.

CREATED  
1997-09-15T07:00:00+0000

COUNTRY  
US

LIVE WHOIS INTELLIGENCE DATA.

Performs domain registration intelligence gathering.  
Displays registrar, creation date, and country information.  
Uses whois-json npm package for backend processing.



# DNS Module

WHOIS  
DOMAIN INTELLIGENCE

DNS  
RECORD LOOKUP

SOCIAL  
USERNAME RECON

WHOIS

DNS

SOCIAL

// DNS RECORD LOOKUP - NAME RESOLUTION INTEL

google.com

A

RESOLVE

| TYPE | NAME       | VALUE           |
|------|------------|-----------------|
| A    | google.com | "142.250.78.46" |

LIVE DNS RESOLUTION RESULTS.

A

ALL

A

MX

TXT

CNAME

NS

Supports A, MX, TXT, NS and CNAME records.  
Displays DNS intelligence information dynamically.  
Uses Node.js DNS module for record resolution.

# Advanced DNS Analysis

WHOIS

DNS

SOCIAL

// DNS RECORD LOOKUP - NAME RESOLUTION INTEL

google.com

MX

RESOLVE

| TYPE | NAME       | VALUE                                                         | TTL |
|------|------------|---------------------------------------------------------------|-----|
| MX   | google.com | {"exchange": "smtp.google.com", "priority": 10, "type": "MX"} | N/A |

LIVE DNS RESOLUTION RESULTS.

MX records provide mail server intelligence.  
TXT records contain verification and security information.  
Useful for reconnaissance and infrastructure analysis.

# Error Handling & Validation

WHOIS  
DOMAIN INTELLIGENCE

DNS  
RECORD LOOKUP

SOCIAL  
USERNAME RECON

WHOIS

DNS

SOCIAL

// WHOIS LOOKUP - DOMAIN REGISTRATION INTEL

google.yzx

LOOKUP

STATUS

No WHOIS data found for this domain

LIVE WHOIS INTELLIGENCE DATA

Handles invalid domains gracefully.  
Displays cyber-themed error messages inside the UI.  
Prevents crashes and improves user experience.

## ▼ OSINT-DASHBOARD

## ▼ client

&lt;&gt; index.html

## ▼ server

&gt; node\_modules

{ } package-lock.json

{ } package.json

JS server.js

# Backend Architecture

Express.js handles API routing.

Frontend communicates using Fetch API.

Backend returns structured JSON responses.

WHOIS and DNS processing occurs server-side.

JS server.js X

server &gt; JS server.js &gt; app.get("/") callback

```
1  const express = require("express");
2  const cors = require("cors");
3  const whois = require("whois-json");
4  const dns = require("dns");
5
6  dns.setServers([
7    "8.8.8.8",
8    "1.1.1.1"
9  ]);
10
11 const dnsPromises = dns.promises;
12
13 const app = express();
14 app.use(cors());
15
16 const PORT = 3000;
17
18 app.get("/", (req, res) => {
19   res.send("OSINT Backend Running");
20 });
21
22
23 app.get("/whois", async (req, res) => {
24
25   try {
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Server running on port 3000

# Workflow / Methodology

User enters domain name

Frontend JavaScript triggers fetch request

Node.js backend receives API request

WHOIS/DNS lookup is processed

JSON response is returned

Frontend dynamically renders results

# Future Scope

- Social OSINT integration
- Sherlock username reconnaissance support
- Threat intelligence integration
- Advanced reconnaissance modules

# Conclusion

- The project successfully integrates WHOIS and DNS intelligence modules.
- Demonstrates practical understanding of APIs and networking.
- Provides hands-on experience in cybersecurity-focused full-stack development.

**Thank You**